

Selenium - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

Official references

<https://www.selenium.dev/>

<https://www.selenium.dev/documentation/>

<https://www.selenium.dev/downloads/>

Prerequisites

Choose the project language and install its runtime/JDK/SDK. For Java configure JDK plus Maven/Gradle; for Python use Python and a virtual environment.

Installation and drivers

Add Selenium through the language package manager. Selenium Manager can resolve drivers in many modern setups; verify browser/driver compatibility when startup fails.

First project

Start WebDriver, navigate, locate elements, perform actions, assert results and close the driver safely.

Troubleshooting

Common issues include driver mismatch, stale elements, synchronization, frames, windows and shadow DOM.

How to start safely

Start from the official Selenium site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

First QA-oriented use

Pick one small, verifiable task in Selenium, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Common issues and troubleshooting

When Selenium fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.